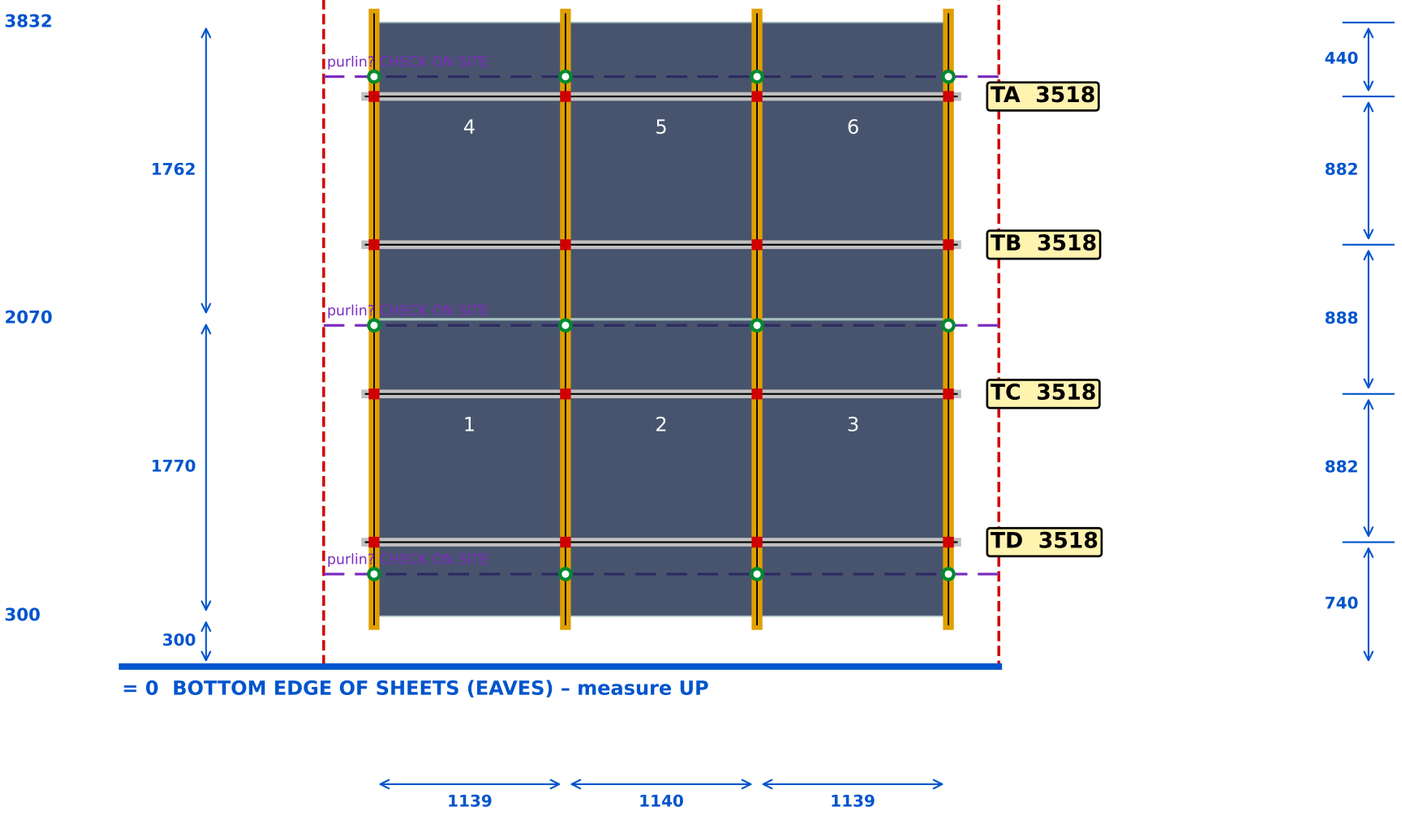
FINAL – 12 PANELS OUTBUILDING – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | OUTBUILDING (12) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v7

2 faces × 6 | PORTRAIT 2 rows × 3 | array 3418 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors

Top of panels 3832 • RIDGE ≥ 4132 – CHECK ON SITE



= 0 BOTTOM EDGE OF SHEETS (EAVES) – measure UP

BASE rails B1…B4 across (centres from left panel edge) – blue mm

Orange = BASE rails UP the roof (3632, B1…B4), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (part – ring Dale to confirm). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 300. Row 1 300–2062, 8 gap, row 2 2070–3832. Base rails 250 → 3882.

Top rails TD 740 TC 1622 TB 2510 TA 3392 (gaps 740, 882, 888, 882, 440)

Slope needed ≥ 4132 vs ~3,800 estimate → 332 SHORT. Width ≥ 4018. CHECK ON SITE

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

NO BIRD MESH ON THIS JOB

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills. Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.**
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571 + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 4 × 3632 up slope (1 bar each, ~1139 apart)

TOP: 4 × 3518 = 3518

Fixings 12 (4 × 3 purlins) Cross conn. 16

End clamps 8 Mid clamps 8 Joiners 0

LOAD LIST – 12 PANELS (both faces)

- ☐ **12 × JA JAM54D40-460/LB**
- ☐ **16 × K2 rail 3.65 m (8 base + 0 full top + 8 for 8×3518)**
- ☐ **0 × joiners ☐ 16 × end 30 mm ☐ 16 × mid 8 mm**
- ☐ **32 × K2 cross connectors – not delivered – ring Dale to confirm**
- ☐ **24 × 2003016 fasteners (have 120 → 96 spare) ✓**
- ☐ **24 × 2002544 plates (have 85 → 61 spare) ✓**
- ☐ **24 × T-bolt 1000571 + 24 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS

1. Count purlins each base rail crosses (3 assumed; 4 → 32 fixings).
2. Slope: needs ≥ 4132 vs est. 3,800 → ~330 SHORT: may not fit 300 mm rule.
3. Cross connectors not on delivery list – ring Dale to confirm before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

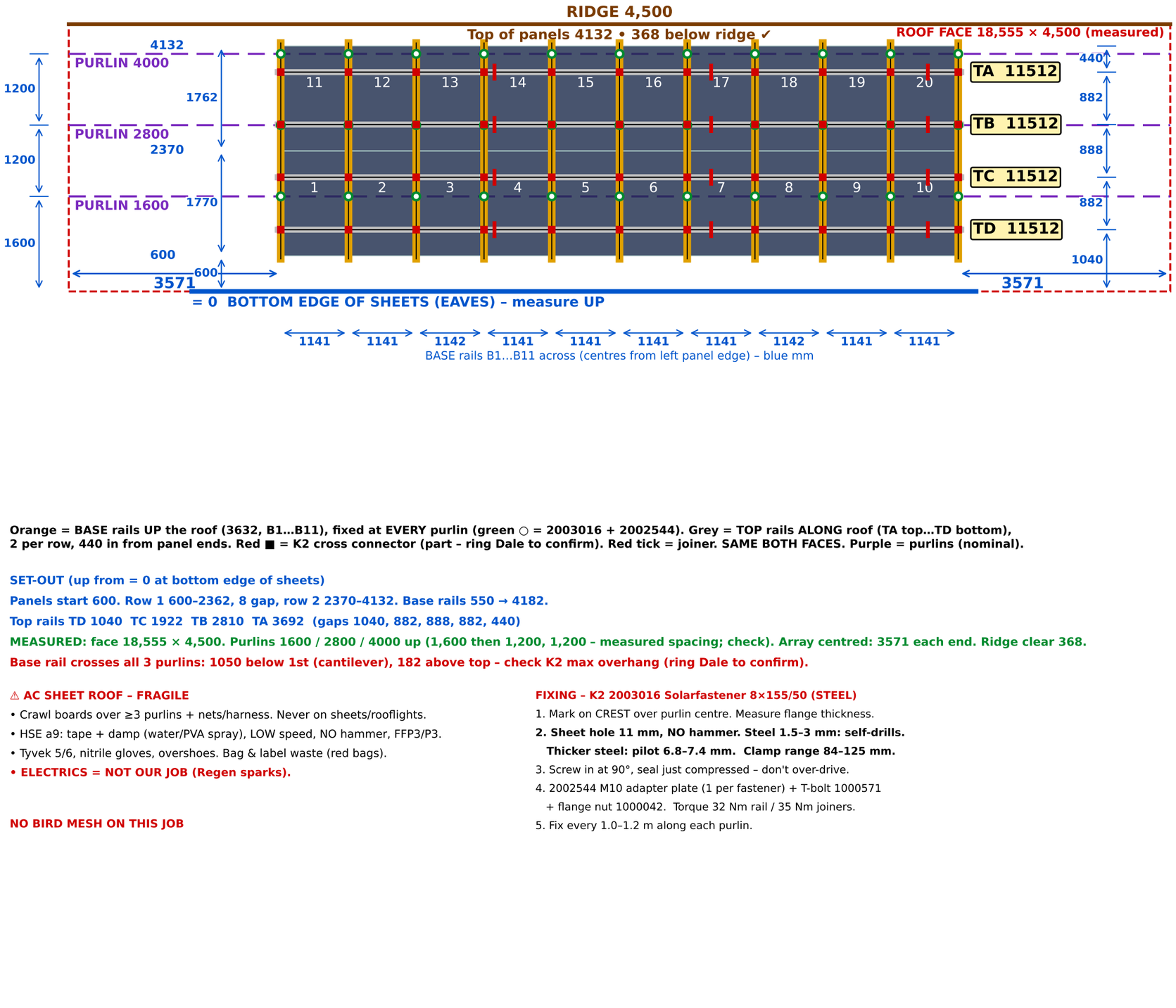
Top-rail cut bars: 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518 | 3518

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL – 40 PANELS MAIN BARN – PORTRAIT ON OVERLAY RAILS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v7

2 faces × 20 | PORTRAIT 2 rows × 10 | array 11412 wide × 3532 up slope | BASE rails UP the roof on purlins, TOP rails ALONG roof on K2 cross connectors



Orange = BASE rails UP the roof (3632, B1…B11), fixed at EVERY purlin (green ○ = 2003016 + 2002544). Grey = TOP rails ALONG roof (TA top…TD bottom), 2 per row, 440 in from panel ends. Red ■ = K2 cross connector (part – ring Dale to confirm). Red tick = joiner. SAME BOTH FACES. Purple = purlins (nominal).

SET-OUT (up from = 0 at bottom edge of sheets)

Panels start 600. Row 1 600–2362, 8 gap, row 2 2370–4132. Base rails 550 → 4182.

Top rails TD 1040 TC 1922 TB 2810 TA 3692 (gaps 1040, 882, 888, 882, 440)

MEASURED: face 18,555 × 4,500. Purlins 1600 / 2800 / 4000 up (1,600 then 1,200, 1,200 – measured spacing; check). Array centred: 3571 each end. Ridge clear 368.

Base rail crosses all 3 purlins: 1050 below 1st (cantilever), 182 above top – check K2 max overhang (ring Dale to confirm).

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

NO BIRD MESH ON THIS JOB

FIXING – K2 2003016 Solarfastener 8×155/50 (STEEL)

1. Mark on CREST over purlin centre. Measure flange thickness.
2. **Sheet hole 11 mm, NO hammer. Steel 1.5–3 mm: self-drills.**
   Thicker steel: pilot 6.8–7.4 mm. Clamp range 84–125 mm.
3. Screw in at 90°, seal just compressed – don't over-drive.
4. 2002544 M10 adapter plate (1 per fastener) + T-bolt 1000571
   + flange nut 1000042. Torque 32 Nm rail / 35 Nm joiners.
5. Fix every 1.0–1.2 m along each purlin.

PER FACE

BASE: 11 × 3632 up slope (1 bar each, ~1141 apart)

TOP: 4 × 11512 = 3650 + 3650 + 3650 + 562

Fixings 33 (11 × 3 purlins) Cross conn. 44

End clamps 8 Mid clamps 36 Joiners 12

LOAD LIST – 40 PANELS (both faces)

- ☐ **40 × JA JAM54D40-460/LB**
- ☐ **48 × K2 rail 3.65 m (22 base + 24 full top + 2 for 8×562)**
- ☐ **24 × joiners ☐ 16 × end 30 mm ☐ 72 × mid 8 mm**
- ☐ **88 × K2 cross connectors – not delivered – ring Dale to confirm**
- ☐ **66 × 2003016 fasteners (have 120 → 54 spare) ✔**
- ☐ **66 × 2002544 plates (have 85 → 19 spare) ✔**
- ☐ **66 × T-bolt 1000571 + 66 × flange nut 1000042**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits
☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm
☐ Hacksaw + 24/32 TPI blades (alu), file/deburr
☐ Tape, chalk line, marker, masking tape, square
☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties
☐ Sealant gun + sealant
☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes
☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes
☐ Crawl boards, harness/lanyards, edge protection
☐ **NO electrical tools – electrics are Regen's**

FLAGS

1. Purlins MEASURED: 3 per base rail – fix at all 3.
2. Slope 4,500 measured – array fits, 368 to ridge, 600 from bottom edge.
3. Cross connectors not on delivery list – ring Dale to confirm before going up.
4. Flange > 3 mm → pilot 6.8–7.4 mm.

Top-rail cut bars: 562+562+562+562+562+562 | 562+562

Not for upload – John's info only. Not a structural design. Purlin-dependent dims marked CHECK ON SITE.

FINAL LOAD LIST + DELIVERED – PORTRAIT OVERLAY – 12 + 40 PANELS

LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | MAIN BARN (40) – AC corrugated sheet on STEEL purlins | Install Wed 4 Nov 2026 | v7

Both systems: base rails UP roof on purlins (40: 3 purlins MEASURED; 12: 3 assumed – CHECK ON SITE), top rails ALONG roof on K2 cross connectors

Item	12-panel	40-panel	TOTAL	DELIVERED (+/–)
JA JAM54D40-460/LB panels	12	40	**52**	**not stated**
K2 rail 3.65 m bars	16	48	**64**	**37 (-27)**
Rail joiners (sleeves)	0	24	**24**	**24 (+0)**
End clamps 30 mm	16	16	**32**	**40 (+8)**
Mid clamps 8 mm	16	72	**88**	**100 (+12)**
K2 cross connectors (ring Dale to confirm)	32	88	**120**	**36 plates?**
2003016 Solarfastener 8×155/50	24	66	**90**	**120 (+30)**
2002544 M10 adapter plates	24	66	**90**	**85 (-5)**
M10 T-bolt 1000571	24	66	**90**	**200 (+110)**
Flange nut 1000042	24	66	**90**	**200 (+110)**
Tigo TS4-A-O optimisers	2	0	**2**	**2 (+0)**
Black plastic clips (end caps?)	?	?	**?**	**? qty**

TOOLS – tick when loaded

☐ Drill/driver + spare batts (NO hammer) ☐ 11 mm + 7 mm bits

☐ Sockets 13/15/17 + T30 ☐ Torque wrench 32/35 Nm

☐ Hacksaw + 24/32 TPI blades (alu), file/deburr

☐ Tape, chalk line, marker, masking tape, square

☐ WIRE CUTTERS ☐ STANLEY KNIFE ☐ Cable ties

☐ Sealant gun + sealant

☐ JSP Force 8 P3 / FFP3, Tyvek, gloves, overshoes

☐ Spray bottle water+PVA, red asbestos bags, labels, OB1 tape, wipes

☐ Crawl boards, harness/lanyards, edge protection

☐ **NO electrical tools – electrics are Regen's**

⚠ AC SHEET ROOF – FRAGILE

- Crawl boards over ≥3 purlins + nets/harness. Never on sheets/rooflights.
- HSE a9: tape + damp (water/PVA spray), LOW speed, NO hammer, FFP3/P3.
- Tyvek 5/6, nitrile gloves, overshoes. Bag & label waste (red bags).
- **ELECTRICS = NOT OUR JOB (Regen sparks).**

SHORTFALLS vs ON-SITE DELIVERY – ring Dale to confirm

- **Rails: need 64 × 3.65 m bars, have 37 → 27 SHORT**
- **Adapter plates 2002544: need 90, have 85 → 5 SHORT**
- **Cross connectors: need 120. '36 metal plates' delivered – CONFIRM: cross connectors or adapter plates? If cross connectors → 84 SHORT**
- **Black plastic joining clips – qty ? (likely rail end caps) – check count**

OK: end clamps 40 (+8), mid 100 (+12), joiners 24 (0), fasteners 120 (+30), T-bolts/nuts 200 (+110), Tigo 2 (0)

NO BIRD MESH ON THIS JOB

Rails: 12-panel top 4 × 3518, base 8 × 3632; 40-panel top 8 × 11512 (3650+3650+3650+562), base 22 × 3632

40-panel: measured 18,555 × 4,500, purlins 1,600/2,800/4,000 – fits ✔

OPTIMISER FITTING – 2 × TIGO TS4-A-O – 12-panel OUTBUILDING

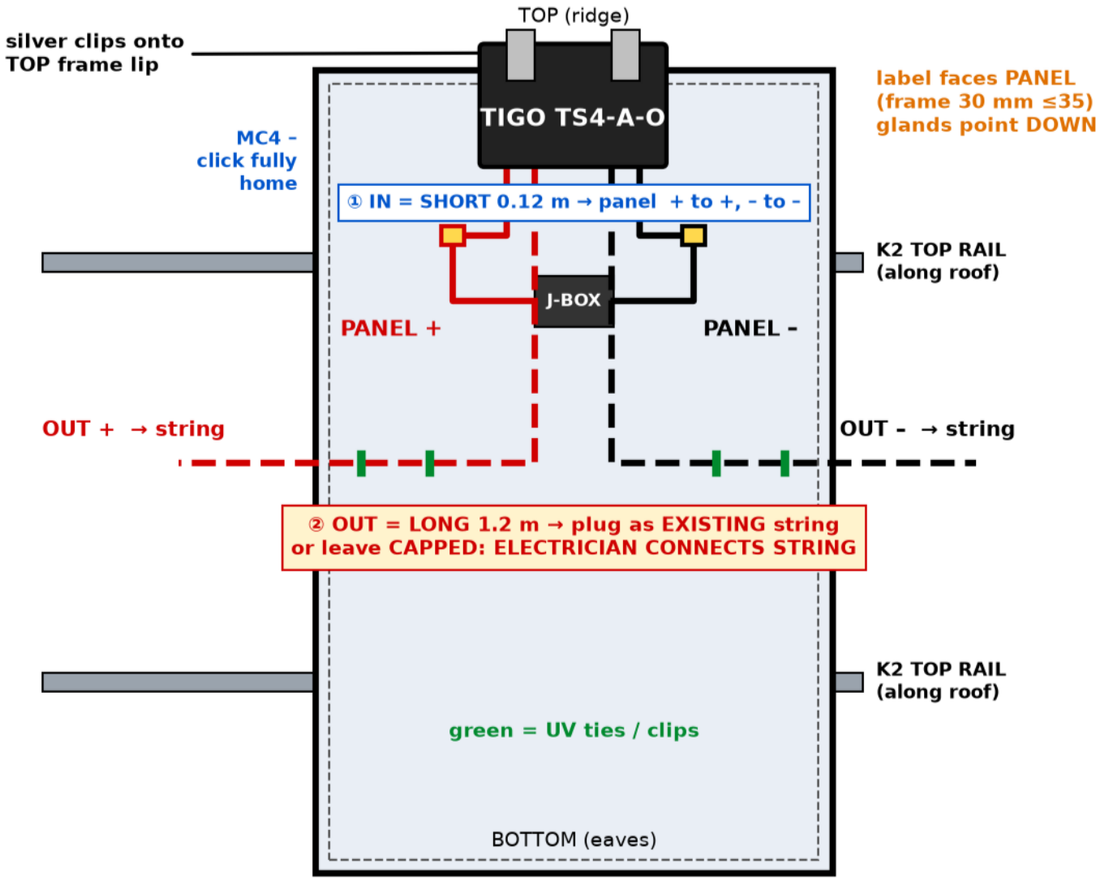
LEWIS – Greenacre Farm, Murcott OX5 2RE | Regen – Dale Collins | JA JAM54D40-460/LB PORTRAIT 1762×1134×30 on K2 overlay rails | v2

PHYSICAL FITTING + PLUG-IN ONLY. No string design, no inverter wiring – ELECTRICIAN CONNECTS THE STRING.

Elizabeth John Designs Ltd – installation sheet
Source: Tigo TS4-A Installation Manual v2.3 (002-00129-00) + TS4-A-O datasheet, tigoenergy.com

REAR VIEW – underside of one portrait panel

(looking at the back, top of panel = ridge end)



Cables tied to rail/frame – NEVER lying on the roof sheet.
No tension, no tight bends, drip loop at each plug.

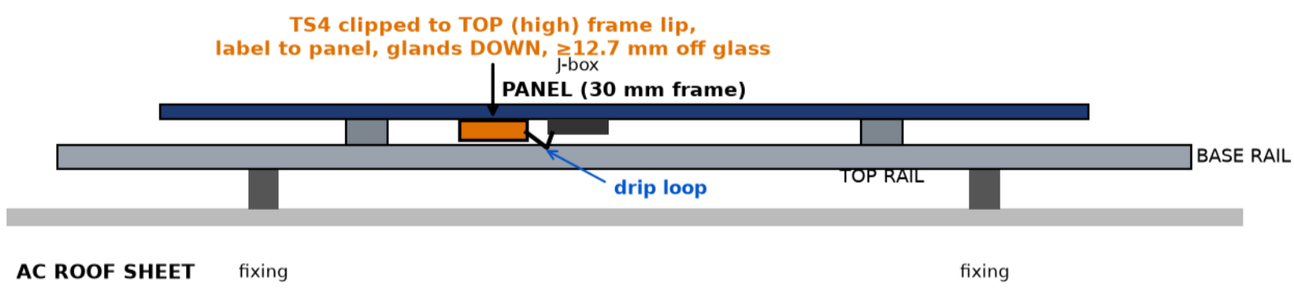


John's photo – Tigo TS4-A-O on site:
- **black box, yellow TS4-A-O label**
- **2 silver frame clips (top edge)**
- **4 MC4 leads: 2 short IN, 2 long OUT**

Guide: tigoenergy.com/documents/installation-manual-ts4-cca-tap

SIDE VIEW – across the slope

TS4, cables, glands & plugs must NOT touch the roof sheet (Tigo).
Keep airflow round it – never sit it in a valley/trough.



TIGO TS4-A-O – KEY FACTS

- Clips onto the TOP (high) edge of the panel frame – silver spring clips
- **JA frame is 30 mm (≤35 mm) → LABEL FACES THE PANEL, glands DOWN**
- No torque on clips. (Frameless only: M8 bolts to rail, 10.2 Nm – not us)
- IN 0.12 m short / OUT 1.2 m long, MC4
- **Ask Regen WHICH 2 panels**

⚠ SAFETY – TIGO WARNINGS

- **Ships switched ON – treat as LIVE**
- **NEVER connect/disconnect under load**
- SHORT IN leads to panel FIRST, then OUT – wrong order can damage it
- Don't mix connector brands / cut MC4
- Plugs out of rain; no jewellery
- Removing: OUT off string first
- Anything odd → STOP, ring sparky

STEPS (from Tigo's manual)

1. Ask Regen which 2 panels. Mark them on the layout sheet.
2. Peel the QR/barcode serial sticker off each TS4 and stick it on the layout sheet at that panel's spot (photo it too).
3. Clip TS4 onto the TOP edge of the panel frame, label facing the panel, cable glands DOWN, before the panel is clamped. Clear of the roof sheet.
4. FIRST plug the SHORT IN leads into the panel: + to +, – to –. Click, tug.
5. THEN the LONG OUT leads: plug on as the existing string, or cap & leave – ELECTRICIAN CONNECTS THE STRING.
6. Tie cables to rail/frame (UV ties), drip loops, no tension, nothing on the sheet. Then clamp the panel down.